Quiz 3 - DSC 10, Fall 2023

Full Name:

PID:

Discussion: ◯ A (3PM) ◯ B (4PM) ◯ C (5PM)

Instructions:

- This quiz consists of 2 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles; if we cannot tell which option(s) you selected, you may lose points. For multiple choice questions, select **only one answer**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- No aids are allowed (no notes, no calculators, and no computers).

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

Ciro is an avid fan of Costco, a members-only warehouse store. On his recent trip to the Morena Boulevard location near campus, he was surprised by how full shoppers' carts were and was inspired to investigate how much people typically spend at Costco.

Question 1

Among all Costco members in San Diego, the average monthly spending in October 2023 was **$350** with a standard deviation of **$40**.

a) The amount Ciro spent at Costco in October 2023 was **-1.5** in standard units. What is this amount in dollars? Give your answer as an integer.

b) What is the minimum possible percentage of San Diego members that spent between **$250** and **$450** in October 2023?

◯ 16% ◯ 22% ◯ 36% ◯ 60% ◯ 78% ◯ 82%

c) Now, suppose we're given that the distribution of monthly spending in October 2023 for all San Diego members is roughly normal. Given this fact, fill in the blanks:

"In October 2023, 95% of San Diego members spent between __(m)__ dollars and __(n)__ dollars."

Give your answers as integers rounded to the nearest multiple of 10.

m: n:

Question 2

Suppose we have access to a simple random sample of all US Costco members of size **145**. Our sample is stored in a DataFrame named `us_sample`, in which the `"Spend"` column contains the October 2023 spending of each sampled member in dollars.

a) Fill in the blanks:

"The average of the `"Spend"` column in `us_sample` is a ____, while the average October 2023 spending of all US members is a ____."

◯ sample statistic; population parameter
◯ population statistic; sample parameter
◯ sample parameter; population statistic
◯ population parameter; sample statistic

b) Fill in the blanks below so that `us_left` and `us_right` are the left and right endpoints of a **46%** confidence interval for the average October 2023 spending of all US members.

```
costco_means = np.array([])
for i in np.arange(5000):
    resampled_spends = __(x)__
    costco_means = np.append(costco_means, resampled_spends.mean())
left = np.percentile(costco_means, __(y)__)
right = np.percentile(costco_means, __(z)__)
```

Which of the following could go in blank (x)? **Select all that apply.**

- ☐ `us_sample.sample(145, replace=True).get("Spend")`
- ☐ `us_sample.sample(145, replace=False).get("Spend")`
- ☐ `np.random.choice(us_sample.get("Spend"), 145)`
- ☐ `np.random.choice(us_sample.get("Spend"), 145, replace=True)`
- ☐ `np.random.choice(us_sample.get("Spend"), 145, replace=False)`
- ☐ None of the above.

What goes in blanks (y) and (z)? Give your answers as integers.

y: _____ z: _____

c) True or False: **46%** of all US members in `us_sample` spent between `us_left` and `us_right` in October 2023.

◯ True ◯ False

d) True or False: If we repeat the code from part (b) 200 times, each time bootstrapping from a new random sample of 145 members drawn from all US members, then about **92** of the intervals we create will contain the average October 2023 spending of all US members.

◯ True ◯ False

e) True or False: If we repeat the code from part (b) 200 times, each time bootstrapping from `us_sample`, then about **92** of the intervals we create will contain the average October 2023 spending of all US members.

◯ True ◯ False

Before submitting your quiz, make sure your PID is on the front page and on the top right corner of page 3.

Fun fact, from Wikipedia: "Costco's earliest predecessor, Price Club, opened its first store on July 12, 1976, on Morena Boulevard in San Diego, California." The company was named after its founder, Sol Price, who also donated $2 million to build UCSD's Price Center!